GENAI-POWERED COLLEGE ADVISOR

A UNIVERSITY SUPPORT NAVIGATOR

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PROBLEM STATEMENT

For a University as large as UNT, it can be extremely difficult for students to find the information that they need outside of scheduling an advising meeting, contacting departments via e-mail or phone, and reading university materials like the Graduate Catalog and scouring different UNT websites.

The main issues can be:

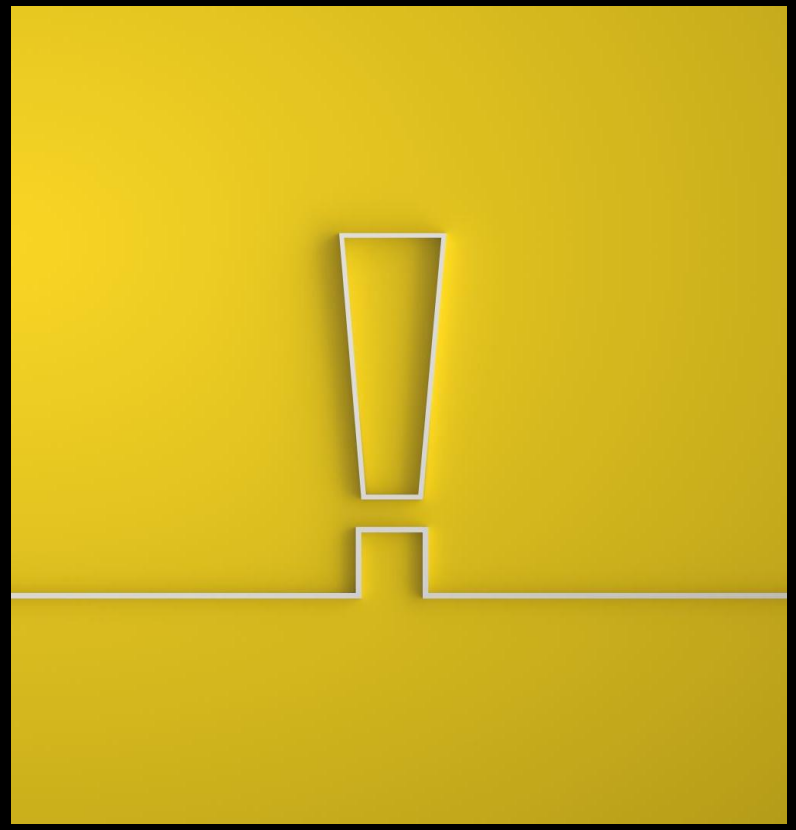
- Lack of availability for Academic Advising, which you may need to schedule weeks in advance.
- Many different departments, divisions, and colleges which all have varying information structures, provided documents/sources, and different questions.
- Time it takes to receive responses via phone or e-mail, and subsequent questions or clarification

THE SOLUTION?

Retrieval Augmented Generation – Large Language Models (RAG-LLMs)

RAG-LLMs can:

- Use a custom input set of documents
- Use different types of embedding models
- Use different types of LLMs
- Be hosted locally
- Have a user-facing, website for accessibility
- Generate text based on the input documents to answer user questions



RAG-LLM METHODOLOGY

1. Obtain data sources.
2. Manually review data sources.
3. Parse Data
4. Create vector embeddings.
5. Store the vector embeddings.
6. Engineer LLM prompt.
7. User queries the system.
8. Query is turned into vector embeddings.
9. Most similar context is returned.
10. Context feeds into the LLM.
11. LLM generates text.
12. Human evaluates text.

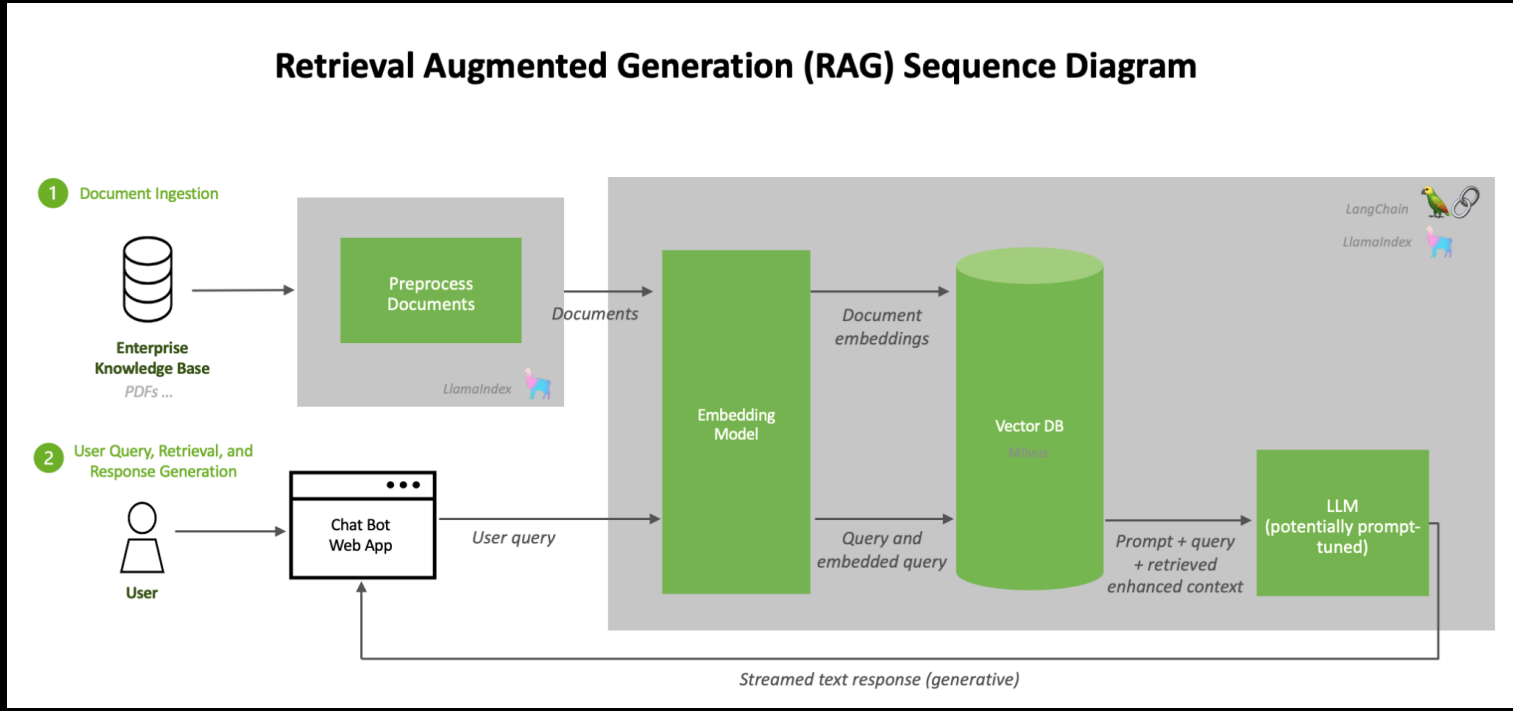


Figure 1: Overview of RAG pipeline components: ingest and query flows [1]

METHODOLOGY

1. Obtain data sources that provide information, answers, or context to help answer questions
2. Review data sources, pre-process documents, or otherwise clean manually.
3. Using a data parser, extract all the text from the various data sources, and create a consolidated structure.
4. Use an embedding model to translate sentences into embedding vectors.
5. Store the vector embeddings in a database for ease of deployment and querying
6. Use the same model to embed user queries, and return similar vector embedding results as context.
7. Use prompt engineering to limit LLM creativity/hallucinations, and act as a specific agent.
8. Feed the most similar context to the LLM model.
9. Have the LLM model generate text given the input context.
10. Compare the results to human results, and have users evaluate.

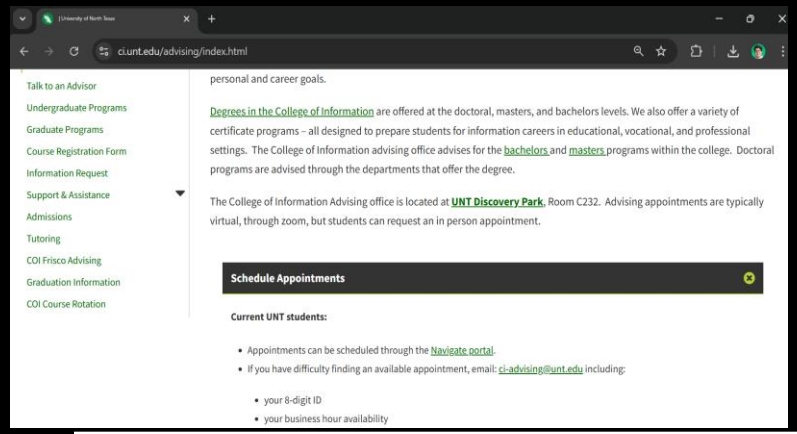
DATA SOURCES

Gathering data can be the most time-consuming part of the process because:

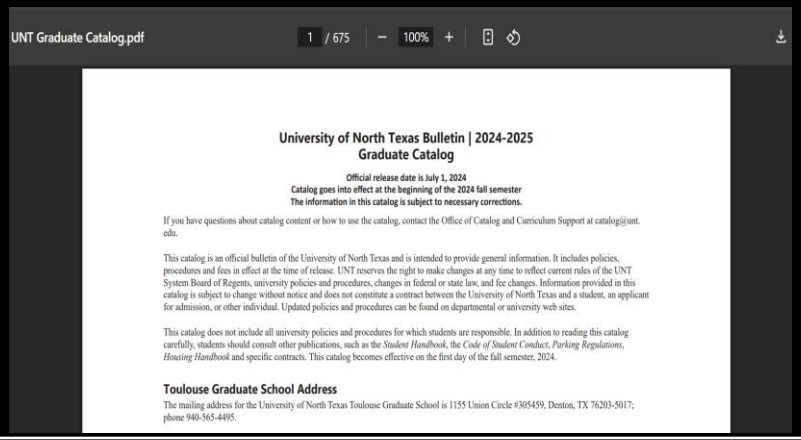
- You must find valid data
- You must clean data
- You must figure out how to extract the data

But how can you approach extracting massive amounts of information from different source types, like:

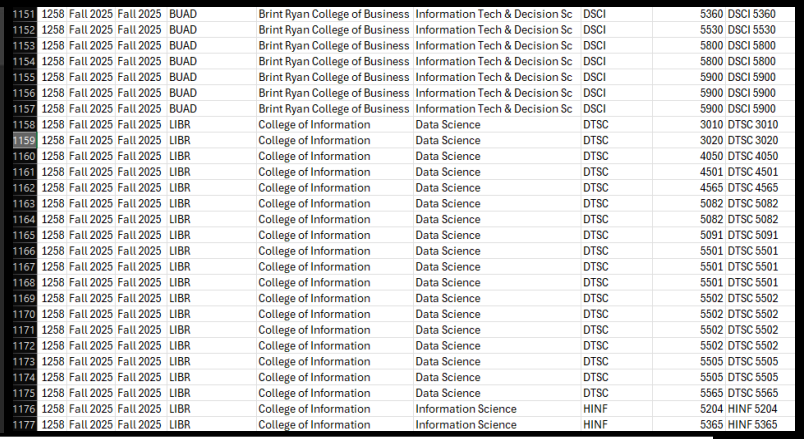
Webpages



PDFs



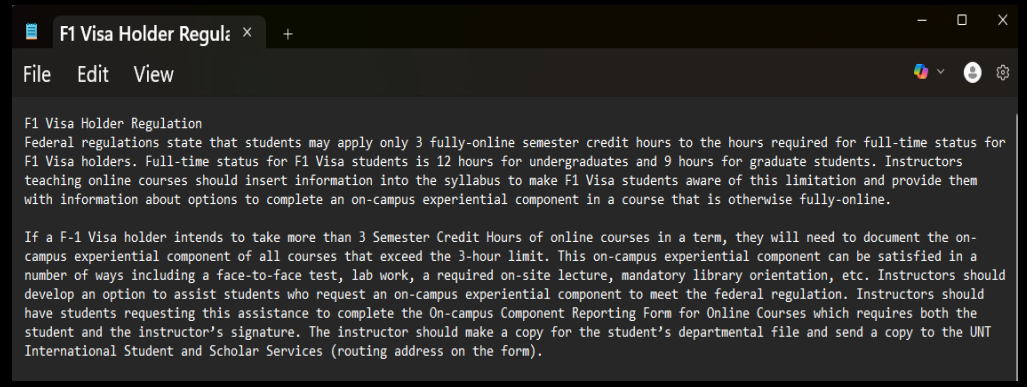
Excel/CSVs



1151	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5360	DSCI 5360
1152	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5530	DSCI 5530
1153	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5800	DSCI 5800
1154	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5800	DSCI 5800
1155	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5900	DSCI 5900
1156	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5900	DSCI 5900
1157	1258	Fall 2025	Fall 2025	BUAD	Brint Ryan College of Business	Information Tech & Decision Sc	DSCI	5900	DSCI 5900
1158	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	3010	DTSC 3010
1159	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	3020	DTSC 3020
1160	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	4050	DTSC 4050
1161	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	4501	DTSC 4501
1162	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	4565	DTSC 4565
1163	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5082	DTSC 5082
1164	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5082	DTSC 5082
1165	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5091	DTSC 5091
1166	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5501	DTSC 5501
1167	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5501	DTSC 5501
1168	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5501	DTSC 5501
1169	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5502	DTSC 5502
1170	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5502	DTSC 5502
1171	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5502	DTSC 5502
1172	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5502	DTSC 5502
1173	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5505	DTSC 5505
1174	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5505	DTSC 5505
1175	1258	Fall 2025	Fall 2025	LIBR	College of Information	Data Science	DTSC	5565	DTSC 5565
1176	1258	Fall 2025	Fall 2025	LIBR	College of Information	Information Science	HINF	5204	HINF 5204
1177	1258	Fall 2025	Fall 2025	LIBR	College of Information	Information Science	HINF	5365	HINF 5365

DATA HANDLING

- Working with Llama Index's SimpleDirectoryReader makes this process a little easier as it can read and parse documents like .pdf, .txt, .json, .pptx, etc.
- However, some of the input files like .pptx may include irrelevant information like slide number, be unable to extract information from images, and in the case of PDF files may have difficulty parsing the data correctly depending on formatting.
- So, while webscrapping, and printing pages as PDF files may work, sometimes to ensure the cleanest information for SimpleDirectoryReader we must copy and paste to .txt, or creating JSON files.



DATA PARSING & PREPROCESSING

Now that we have a set of documents that look to answer our College of Information Advising questions, we can use SimpleDirectoryReader, which will parse all the metadata and texts and store as a Document type.

We can then perform our normal text pre-processing via code such as making all text lowercase, removing multiple line breaks, removing unnecessary special characters, multiple white spaces, and multiple periods in a row.

- DTSC 5501 - Fundamentals of Data Analytics
- DTSC 5502 - Principles and Techniques for Data
- DTSC 5505 - Applied Machine Learning for Data
- Dual_Credit_FAQ
- Enrollment Holds
- F1 Immigration Regulations
- F1 International Student Employment
- F1 Students I20 Request through iNorthTX
- F1 Visa Holder Regulation
- fall-2025-academic-calendar
- Financial_Aid_FAQ
- First_Flight_FAQ
- Good Neighbor Program
- How to change your major
- How to Maintain F1 Status

```
/content/College_Of_Information_Chatbot/2024-2025 Academic Calendar by Semester.pdf
5
Spring 2025 Calendar
KEY SEMESTER DATES
Full
Semester
Jan. 13 -
May 9
3 Week
I Winter
Session
Dec. 16 -
Jan. 10
8 Week I
Session
Jan. 13 -
Mar. 7
8 Week II
Session
Mar. 17 -
May 9
Schedule of Classes Available on myUNT Sep 23,
2024
```

```
Document 0: 4 2024-2025 academic calendar dates are subject to change by official action of unt. fa

Document 1: 5 spring 2025 calendar key semester dates full semester jan. 13 - may 9 3 week i winter

Document 2: 6 summer 2025 calendar key semester dates 3 week i session may 12 - may 30 5 week i ses

Document 3: data analytics ii 2024-2025 graduate catalog adta 5230 - data analytics ii 3 hours exte

Document 4: admission process for masters in data science. we are currently accepting applications

Document 5: apply for degree conferral graduation commencement eligibility and applying for degree

Document 6: biomedical engineering laboratory2024-2025 graduate catalog bmen 5210 - biomedical engi

Document 7: computational methods in biomedical engineering 2024-2025 graduate catalog bmen 5315 -

Document 8: biomedical engineering seminar2024-2025 graduate catalog bmen 5940 - biomedical enginee

Document 9: information retrieval and web search 2024-2025 graduate catalog csce 5200 - information

Document 10: modeling and simulation 2024-2025 graduate catalog csce 5213 - modeling and simulation
```

TEXT SPLITTING & CHUNKING

Text splitting is the process of using a criteria to break apart each document's text into chunks of specific lengths (N).

- Character Splitting breaks text into chunks of length N characters.
- Sentence splitting breaks text into chunks of length N sentences.
- More advanced types like Recursive Character Splitter or Semantic Splitter look to break text into chunks of length N, based online breaks, or sentence semantics, respectively.

For our model we chose to use SentenceSplitter in the Llama Index library, with a chunk size of 512 and an overlap of 50. By using large chunks with a large overlap, we attempt to contain a lot of context within each chunk to keep contextual information and meaning together.

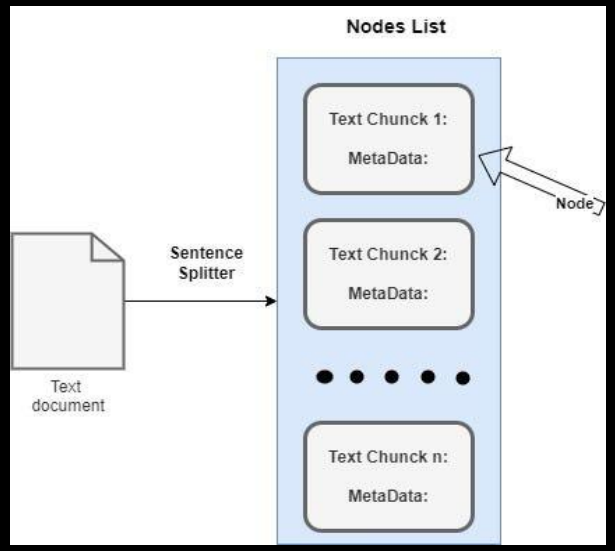


Figure 2: SentenceSplitter splitting text document into multiple chunks stored in nodes [2]

Text splitting and chunking methodologies has the ability to be a whole area of research.

TEXT EMBEDDING

- Text embedding is the process of using a model to translate each word in a chunk to an array of numbers which can be used to determine distance and similarity between each other.
- For our project we are using the framework SentenceTransformers with the embedding model BAAI/bge-m3.
- Text embedding is used to create embeddings of the data documents, and the user entered question.

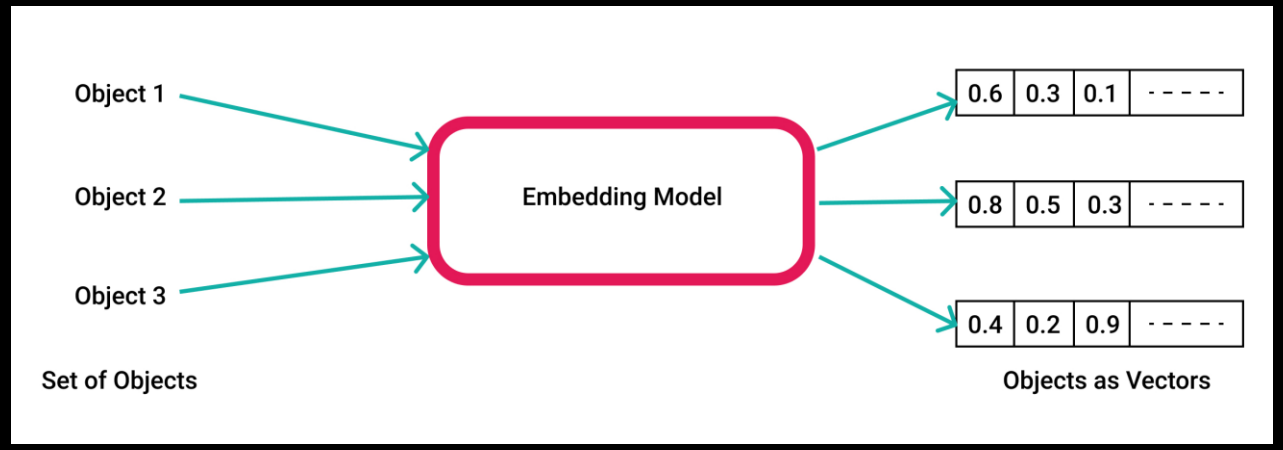


Figure 3: Text embedding model workflow [3]

VECTOR STORAGE

Once the vector embeddings are created, they will be stored in a database for querying later. For our project we used ChromaDB an open-source embedding database.

This allows the comparison between user query and vector embeddings of documents to be more structured.

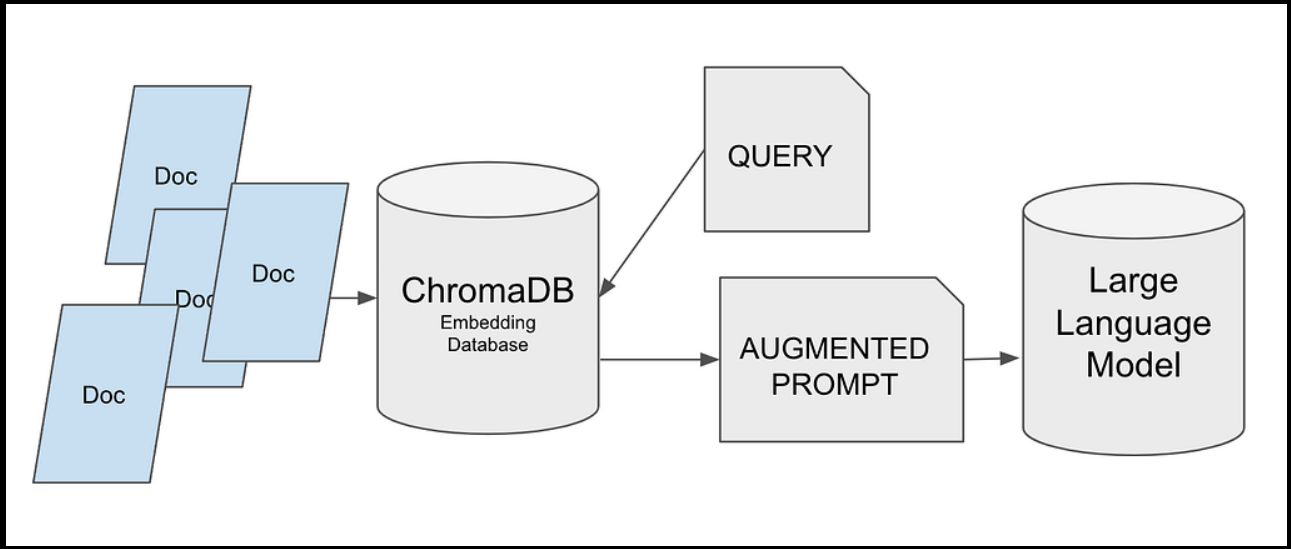


Figure 4: ChromaDB workflow [4]

CONNECTING TO A LLM

One of the hardest parts of implementation is getting a connection to a LLM model.

- LLM models are computationally taxing when running locally
- APIs to the largest, newest, and best models can be cost-prohibitive

For running the model, Christian had a desktop with:

- 64Gb RAM
- RTX 4090 24Gb VRAM

For our project we used Meta's Llama 3.3 70b.

- Model has 70 billion parameters
- Model size is 42Gb

PROMPT ENGINEERING

Engineering a prompt is important for working with LLMs, especially RAG-LLMS, because:

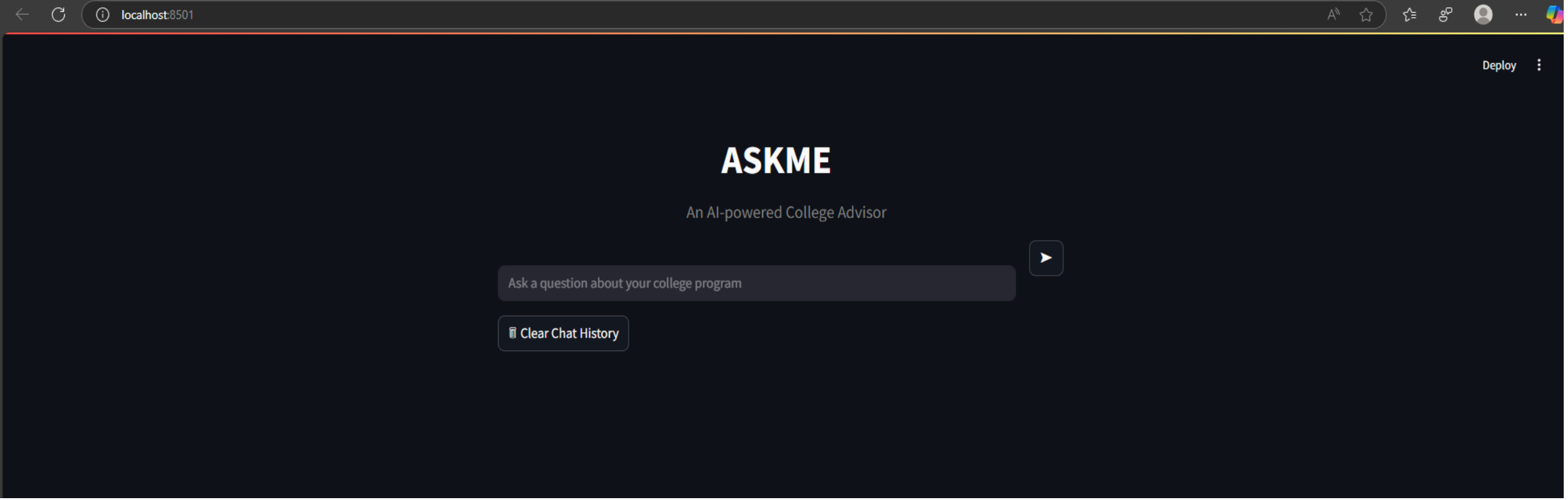
- You want to limit the model using pre-trained information outside of the document set you give it
- You want it to act as a particular agent type, like a UNT Academic Advisor
- You would like to receive specific formats for output data

```
#Prompt Engineering using Context and Chat History
prompt = f"""
You are an academic advisor for the University of North Texas College of Information.
You specialize in helping students in the Masters in Data Science program.
Only use the provided context and chat history. Do not guess or fabricate answers.
Include relevant phone numbers, office locations, e-mails, or website links.

Context:
{context}

Chat History:
{history_str}

Question: {query}
Answer:"""
```



USER INTERFACE

- Streamlit provides an easy to program website interface in python. While our project is only available locally, Streamlit also provides options to publish websites for distribution.
- We created a User interface using Streamlit so that users do not need to program each individual question but instead can enter the question via a user-friendly text box, and can see the answer streamed out, time it took to process, and get information about the sources it came from for human evaluation.

HUMAN EVALUATION

- LLM model output can be hard to evaluate because the output is not a number or classification that can use common metrics like accuracy, f1-score, or Mean Squared Error.
- Instead, one option is to use human evaluations, where a human reviews the output of the model to judge how well the LLM performed.
- We also obtained baseline answers from a College of Information academic advisor to our questions to compare generated text to.
- Relevance – Is the response related to the user's question?
- Correctness – Is the answer accurate and supported by retrieved context?
- Completeness – Does the response cover all the important aspects of the query?
- Fluency – Is the language clear, grammatical, and natural?

EVIDENTLY AI

Using an AI tool to evaluate some of these metrics and others can speed up the process of evaluations, but can leave reasoning up to another automated model, which may cause errors!

Using Evidently AI we were able to calculate:

- Correctness – Compare model response to human given response
- Context Quality – Does the context provide information to answer the question?
- Faithfulness – Does response stay faithful to context?
- Text similarity metrics – are the response words similar in meaning to the human response?
  - BERT Score & Semantic Similarity

RESULTS

Low because..?

Human responses were:

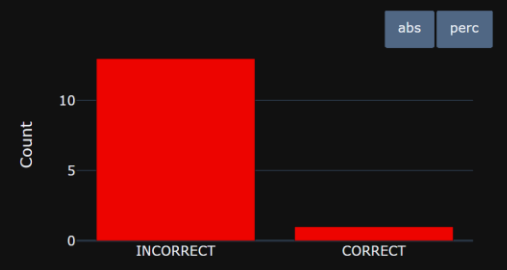
- Short
- In a condensed time, frame
- More about getting general info
- Dictated

High because..?

Our context was able to answer the question in most cases

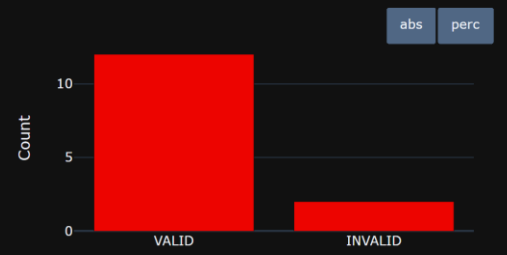
Correctness_1
cat

	current
count	14.00
missing	0.00 (0.00%)
INCORRECT	13.0 (93%)
CORRECT	1.0 (7%)



ContextQuality
cat

	current
count	14.00
missing	0.00 (0.00%)
VALID	12.0 (86%)
INVALID	2.0 (14%)

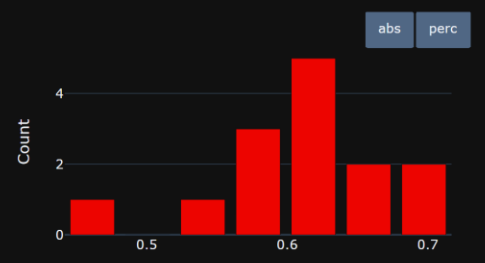


RESULTS

Overall, above average score. Possibly lower than semantic similarity because BERT is looking for token-based similarity and can be sensitive to different phrasing.

BERTScore
num

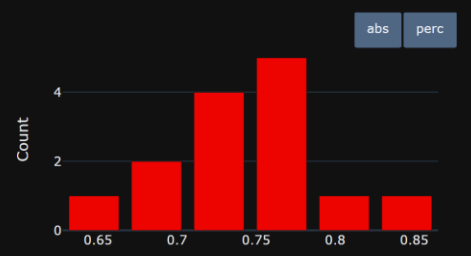
	current
count	14.00
missing	0.00 (0.00%)
min	0.46
max	0.74
mean	0.63
std	0.07



Overall above average, possibly because it focuses on sentence embedding to convey intent or meaning. It can be skewed by vague or superficially similar sentences though.

Semantic Similarity
num

	current
count	14.00
missing	0.00 (0.00%)
min	0.65
max	0.88
mean	0.77
std	0.06



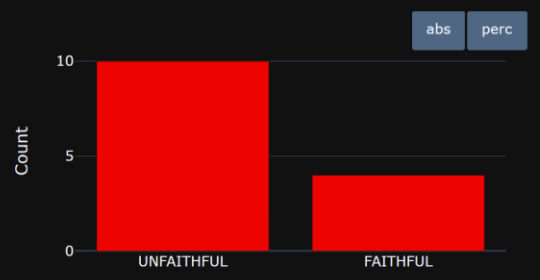
RESULTS

Overall majority of responses were considered unfaithful, which means the generated responses may contain hallucinations and other information that was not provided in the context.

This could also be exacerbated by including chat history in the Prompt Engineering, as chat history was not appended to the context before evaluating for faithfulness.

Faithfulness_1
cat

	current
count	14.00
missing	0.00 (0.00%)
UNFAITHFUL	10.0 (71%)
FAITHFUL	4.0 (29%)



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count	14.00
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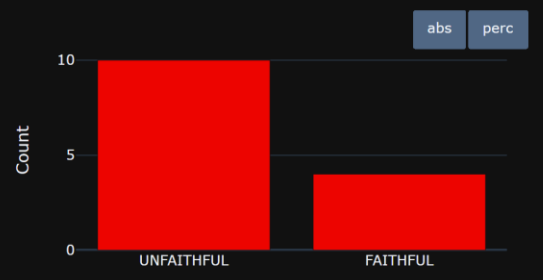
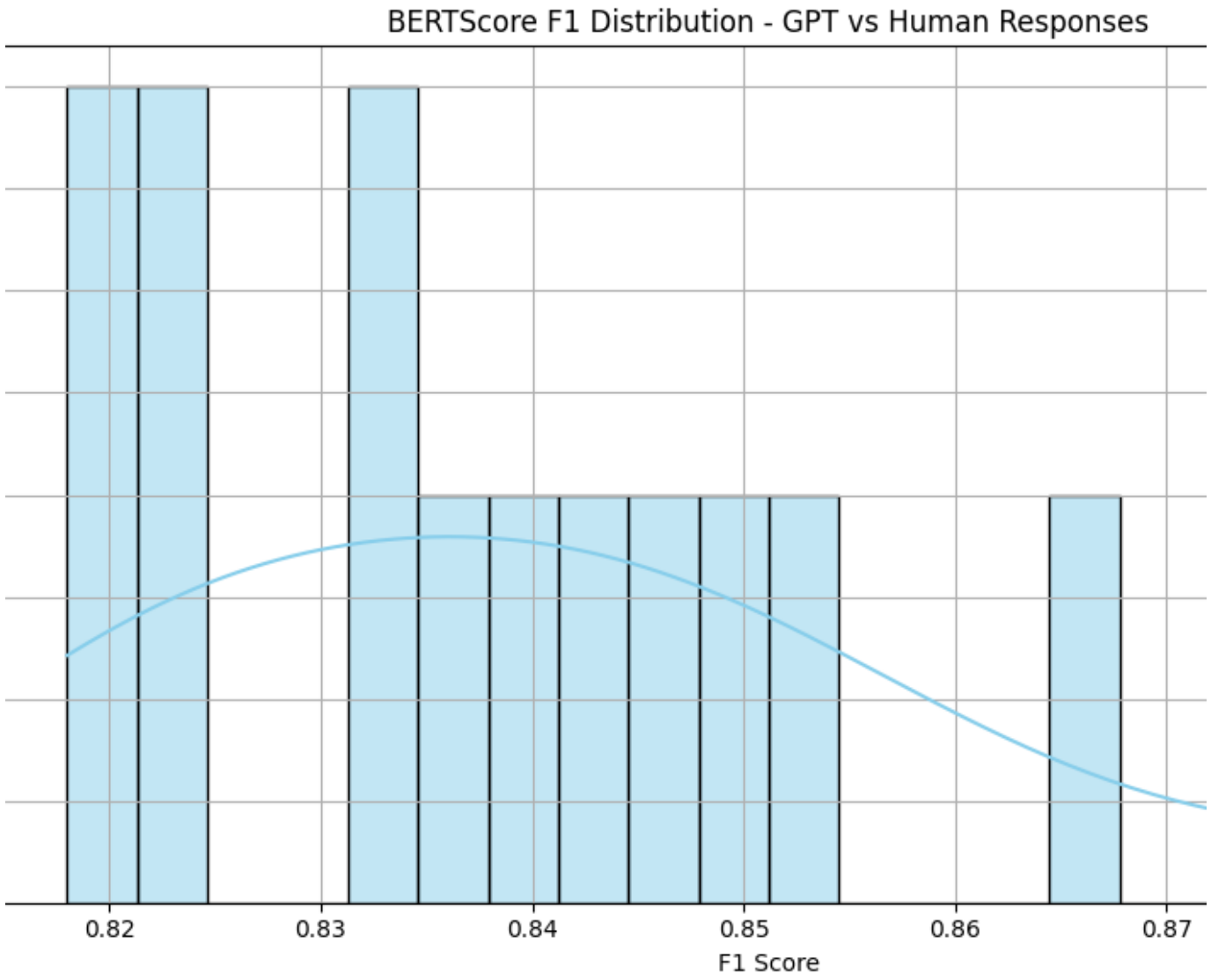


RESULTS

BERTScore F1 values between 0.82 and 0.88 are generally considered strong

- GPT responses show **strong semantic similarity to human responses**
- Slight variance is natural and shows room for optimization
- Most scores lie between **0.82 and 0.85**



RESULTS

- **Majority (11 out of 14)** responses were labeled as **"Possible Hallucination"** — this indicates a significant **factuality issue** in the generated outputs.
- Only **3 responses** were **"Partially Factual"**, which shows that even fewer responses met the standard of mixed accuracy.
- No responses are labeled as fully factual. So the results can be improved by context-awareness, Fine-tuning and prompt engineering

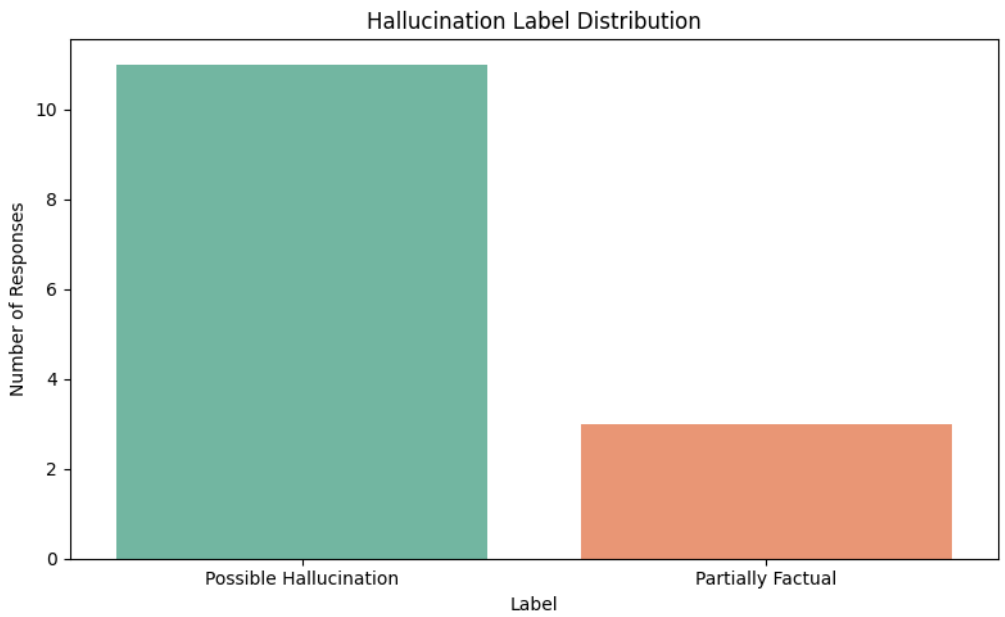


Fig: Hallucination Label Distribution

RESULTS

- **Reranking module is clearly improving the contextual relevance** (cosine similarity).
- **Rerank performs significantly better** on average and shows less variance
- Vector-based retrieval has more outliers and lower median similarity
- **BERTScore remains consistently high**, reranking helps **better align the answer with the query**
- As the response length increases, the BERTScore F1 tends to decrease

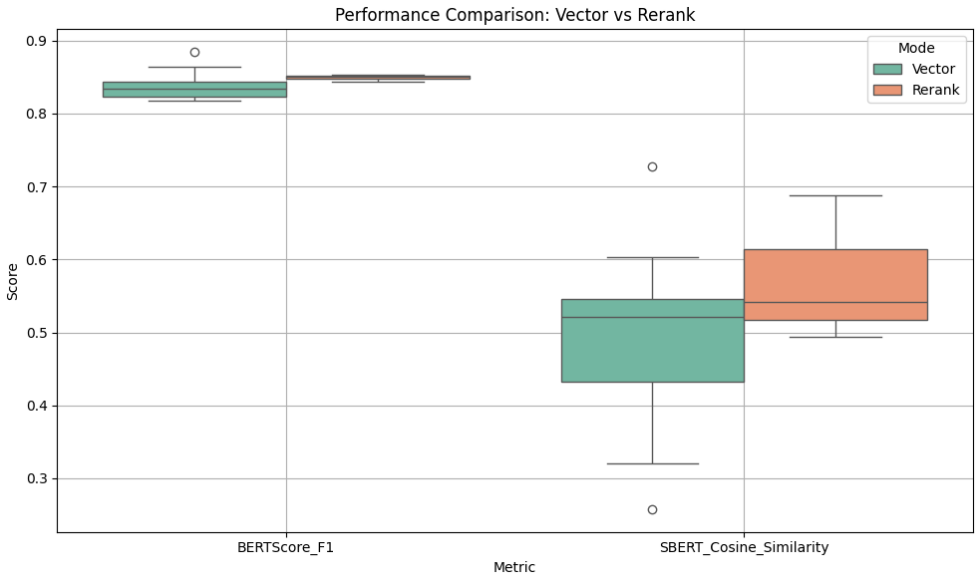


Fig: Performance Comparison — Vector vs Rerank

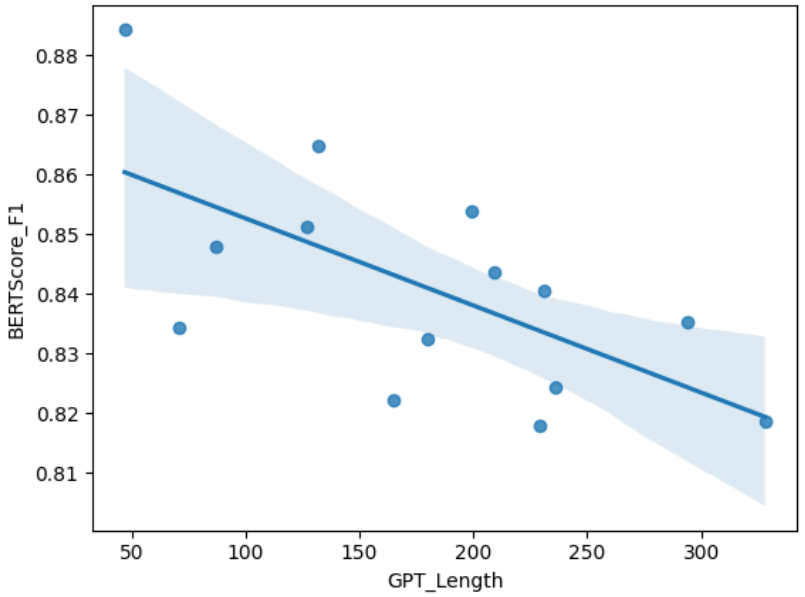


Fig: BERTScore F1 vs GPT Response Length

RESULTS

Ask your question:

How do I remove a hold from my account?

Advisor Response: "This depends on the hold, and which department placed the hold. For instance, if the hold is placed by student accounting, the student will need to contact the student accounting office. Sometimes holds can be released by the student by completing something in their my.unt.edu, however if not, contact the department."

Evidently AI Correctness: Incorrect, The OUTPUT introduces additional claims and step-by-step instructions on how to remove a hold that are not present in the REFERENCE. The REFERENCE specifically mentions contacting the relevant department, particularly student accounting, and that some holds can be released by the student through my.unt.edu. The OUTPUT, however, elaborates further on identifying types of holds, reasons for them, and potential actions to take, which deviates from the original meaning and focus of the REFERENCE.

BERT Score = 0.655
Semantic Similarity = 0.80

Answer

To remove a hold from your account, you will typically need to take specific actions or provide required documentation to address the reason for the hold. Holds can be placed on your account for various reasons such as unpaid fees, missing documents, academic issues, or other administrative requirements. Here are general steps you might follow:

1. **Identify the Type of Hold**: First, determine the type of hold that has been placed on your account. This information is usually available through your university's online portal or by contacting the registrar's office or student services.
2. **Understand the Reason for the Hold**: Once you know what type of hold it is, find out why it was placed on your account. Common reasons include outstanding balances, incomplete admission processes, health insurance requirements, or academic probation.
3. **Take the Required Action**: Based on the reason for the hold, take the necessary steps to resolve the issue. This might involve:
   - Paying an outstanding balance.
   - Submitting required documents (e.g., transcripts, vaccination records).
   - Meeting with an academic advisor to discuss your course schedule or academic performance.
   - Providing proof of health insurance.
4. **Confirm the Hold Has Been Removed**: After you have taken the necessary action, confirm with the relevant department that the hold has been removed from your account. This might involve checking your online portal again or contacting the office directly.
5. **Follow Up**: If the hold was preventing you from registering for classes or accessing other university services, follow up to ensure you can now complete these tasks without issue.

For specific guidance tailored to your situation, it's best to contact your university's support services directly. They can provide detailed instructions based on their policies and procedures. For students at the University of North Texas (UNT), you can reach out to the appropriate office such as the Registrar's Office or the Student Accounting and Cashiering Services for assistance with removing holds related to registration, financial obligations, or other administrative matters.

ASKME

An AI-powered College Advisor

How do I remove a hold from my account?

Final Answer

To remove a hold from your account at the University of North Texas (UNT), you should follow these steps:

1. Log in to your my.unt.edu portal to identify the specific holds on your account.
2. Click on each hold listed to get more detailed information about the nature of the hold and the steps required to resolve it.
3. Follow the instructions provided for each hold. These may vary depending on the type of hold. For example:
   - If you have an Official Transcript hold, you can either bring your official academic documents to the Admission's Desk on the 1st floor of the Eagle Student Services Center or have your school directly email your transcripts to GradAdmission@unt.edu or UnderGrad.Admissions@unt.edu, depending on whether you are a graduate or undergraduate student.
   - If you have a TB block due to the screening process, the Student Health and Wellness Center (SHWC) staff will remove this hold from your account once the screening is complete.
   - For international students, specific information regarding holds related to the Intensive English Language Institute (IELI), Optional Practical Training (OPT) programs, or general international student insurance can be found on the respective informational flyers.

Remember that this is not a complete list of possible holds, so it's essential to check your my.unt.edu portal for the exact holds on your account and the specific steps to resolve them. If you need further assistance or clarification, you can contact the appropriate department or office at UNT that is responsible for the hold.

Response Time: 8.90 seconds

View Context Chunks

Previous Q&A

Clear Chat History

RESULTS

Ask your question:

How do I remove a hold from my account?

Context Quality: Valid.

The text provides clear instructions on how to view and address holds on an account by detailing the steps to log in, check for holds, and contact the relevant offices to remove them. It mentions specific types of holds and who can resolve them, effectively answering the question.

Faithfulness: Unfaithful.

The response provides general steps for removing a hold and adds specific actions that may not be explicitly mentioned in the source, such as providing proof of health insurance and meeting with an academic advisor. Additionally, it does not cite the specific types of holds as outlined in the source, which includes 'advising required', 'UNT registration guide', 'past due balance', and 'missing final transcript'. The response also suggests contacting the Registrar's Office or Student Accounting and Cashiering Services, which while potentially helpful, is not grounded in the specific context provided in the source regarding holds and their resolutions.

RESULTS

Ask your question:

How do I get admitted to the Master's in Data Science program in the College of Information?

Advisor response: "https://datascience.unt.edu/programs/msds/admission.html Minimum GPA, background in math and programming. Leveling up courses for admission requirements."

Answer

To get admitted to the Master's in Data Science program in the College of Information, you can follow these steps:

1. Visit the department's website at http://informationscience.unt.edu to access application forms and current program information.
2. Review the admission requirements, which may include a bachelor's degree from an accredited institution, GPA requirements, and other prerequisites.
3. Submit your application, transcripts, and other required documents to the College of Information's advising and admissions office.
4. Contact the department's academic advisor or admissions office for guidance on the application process and to discuss any specific requirements or questions you may have.

You can also contact the College of Information's advising and admissions office directly at:

- Phone: 940-565-2445
- Email: ci-advising@unt.edu
- Location: Discovery Park, Room C232

Additionally, you can check the university's website for more information on the Master's in Data Science program, including course offerings, degree requirements, and career opportunities.

Response Time: 136.73 seconds

ASKME

An AI-powered College Advisor

How do I get admitted to the Master's in Data Science program in the College of Information?

Final Answer

To get admitted to the Master's in Data Science program in the College of Information, you will need to follow these steps:

1. Ensure you meet the admission requirements:
   - You should be a major in the UNT information science Bachelor of Science program if you are considering the grad track option.
   - Be prepared for a program that is mathematics and programming intensive.
2. Prepare your application materials:
   - Write an experience write-up detailing your background and why you are interested in the program.
   - Update your resume or CV to reflect your academic and professional experiences.
   - Obtain two (2) recommendations from individuals who can speak to your qualifications for the program.
3. Start your application:
   - If you are applying for the Summer 2025, Fall 2025, or later terms, go to the following link to start your application: https://unt2025.cas.myliaison.com/applicant-ux/#/login
   - Follow the detailed instructions for the application process provided on the UNT Graduate School admissions page: https://tgs.unt.edu/future-students/graduate-admissions/how-to-apply-liaison.html
4. Submit your application:
   - Complete the Data Science application form as directed on the College of Information, Department of Information Data Science application portal: https://informationscience.unt.edu/admission-ms-data-science
   - Include all the required documents and information as part of your application package.
5. Await the admission decision:
   - After submitting your application, the admissions committee will review your materials and make a decision. You will be notified of the outcome.

Remember to check the specific deadlines for the term you are applying for and ensure that you submit all materials on time. If you need further assistance or have specific questions, you may contact the

RESULTS

Ask your question:

How do I get admitted to the Master's in Data Science program in the College of Information?

Advisor response: "https://datascience.unt.edu/programs/msds/admission.html Minimum GPA, background in math and programming. Leveling up courses for admission requirements."

Correctness: Incorrect
The OUTPUT introduces additional details about specific application steps, such as requiring letters of recommendation, entrance exams, and detailed descriptions of the program's structure and content. The REFERENCE only mentions a minimum GPA and a background in math and programming as parts of the admission requirements, which the OUTPUT does not directly reflect. Therefore, the OUTPUT contains information not present in the REFERENCE, altering the original meaning.

BERT Score = 0.64
Semantic Similarity = 0.76

Context Quality: Valid
The text clearly outlines the admissions process for the Master's in Data Science program, including application deadlines, GPA requirements, required documents, and the process for submitting transcripts and letters of recommendation. It also provides information on how to apply, what materials are necessary, and mentions that GRE or GMAT scores are not required, which are all essential details for answering the question about admission.

Faithfulness: Unfaithful
The response contradicts the information provided in the source. Specifically, it mentions entrance exams (GRE or other) being required, but the source explicitly states that GRE or GMAT scores are not required for admission to the program. Additionally, it does not accurately reflect the specifics of the application steps and document requirements as outlined in the source.

RESULTS

Some sample questions & responses.

I have work at the Union. Where should I park as a visitor?

Final Answer

As a visitor with work at the University Union, you should park at Lot 49, which is located at Union Circle in front of the Union. This lot is designated for visitors and offers paid parking. It is also noted to have accessible parking available, should you require it, and there are no restricted hours mentioned for this lot, so you can park there at any time. The Union Circle Garage is also an option for hourly parking, and it

available food items at food pantry?

Final Answer

At the food pantry, a variety of food items are available to cater to different dietary needs. These include protein options such as chicken, tuna, salmon, beans, and vegetables. There are also gluten-free and gluten-sensitive options. For those with specific dietary requirements, the pantry offers items suitable for vegetarian, vegan, gluten-free, lactose-intolerant, pescatarian, and kosher diets. Additionally, the Freedom Food Pantry provides fruits, vegetables, desserts, breads, and meats. Students can request a variety of food that may serve their dietary needs, and if certain items are not available, they can email DESresources@unt.edu to request additional items be added to the rotation. It's important to note that students must bring their UNT ID Card to utilize the food pantry.

Response Time: 5.81 seconds

Can a master's student enroll in an undergraduate course?

Final Answer

Based on the provided context, a master's student may not receive graduate credit for any undergraduate course by taking the course under a 5000- or 6000-level designation, such as special problems. Undergraduate courses, except those which meet with graduate courses, are not typically part of the master's degree curriculum. Therefore, while a master's student could technically enroll in an undergraduate course, they would not receive graduate credit for it unless it is one of the courses that meet with graduate courses. It is always advisable for students to consult with their major professor and advisory committee to ensure that any courses they take align with their degree plan and program requirements.

Response Time: 4.32 seconds

Final Answer

The food pantry hours at the University of North Texas are as follows:

- Friday: 10am-5pm
- Saturday: 10am-3pm
- Sunday: CLOSED

Please note that these hours are specific to the UNT food pantry. If you need further assistance or information, you can visit the UNT food pantry website or contact them directly.

Response Time: 3.16 seconds

I have work at chestnut hall. Best place to park my car.

Final Answer

Based on the context provided, if you are a student using services at the SHWC (Student Health and Wellness Center), you should park in the white Patient Parking spaces in Lot 5. Remember to request a parking validation at the time of check-out, as the SHWC will not issue validations for patients who park in metered spaces. If you are not using SHWC services and just need to park for work at Chestnut Hall, you will need to find alternative parking. You may consider parking in any available general parking spaces that you are permitted to use according to your parking pass or permit. If you do not have a parking pass, you may need to use metered parking or find street parking nearby, keeping in mind the parking regulations and potential need for payment. Always make sure to follow the university's parking rules to avoid citations or towing.

Response Time: 4.45 seconds

Who should I contact for transportation-related issues?

Final Answer

For transportation-related issues at the University of North Texas, you should contact Transportation Services. You can reach them by phone at 940-565-3020 or by email at transportation.services@unt.edu. If you need assistance during office hours, their office is located on the UNT campus on the bottom floor of the Highland Street Garage at 621 Avenue A, Denton, TX 76201.

Response Time: 2.72 seconds

What are the core courses in the MS in Data Science?

Final Answer

The core courses in the MS in Data Science program, as outlined in the provided context, are:

- INFO 5501 or DTSC 5501 - Fundamentals of Data Analytics
- INFO 5502 or DTSC 5502 - Principles and Techniques for Data Science
- INFO 5505 - Applied Machine Learning for Data Scientists

These core courses constitute 9 credit hours of the required 36 credit hours for the Master of Science in Data Science degree program.

Response Time: 3.02 seconds

MORE METRICS

Time taken by both models to generate 14 questions.

Model	Time (Sec)	Average Time (Sec)
Ollama	2927.52	209.11 seconds
OpenAI (GPT-4o)	86.3	6.16

For comparison human advising appointments are usually only 30 minutes but must be scheduled weeks in advance (depending on time of year). In my advising appointment, the advisor was able to provide general answers and links for 14 questions in about 20 minutes.

FUTURE CONSIDERATIONS

- Agentic AI for Task Completion
- Human-Like Evaluation System
- Smarter Data Handling
- Fine-Tuning the Model
- Expanded Dataset Coverage
- User Interface, that can handle multiple requests from students and quickly respond.

REFERENCES

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[2] Samad, Abdul (2024, May 20). *Agentic RAG Application using LlamaIndex – Router Query Engine.* Medium. https://medium.com/@samad19472002/agentic-rag-application-using-llamaindex-router-query-engine-5b3f7b7feb75

[3] Tripathi, Rajat (2023, Jun. 30). *What are Vector Embeddings*. Pinecone. https://www.pinecone.io/learn/vector-embeddings/

[4] Bhoi, Debaprasann (2024, Feb. 11). *ChromaDB: An Open-source vector database*. Medium. https://blog.gopenai.com/chromadb-an-open-source-vector-database-f1871580b412